Sensitivity, Robustness and Loop Shaping

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Where this fits:

- **Builds on:** Block-diagram algebra; Bode plot; Gain, phase and delay margins.

1. Overview

Key point: How close the heading loop comes to the critical point -1 is one number, the peak of the sensitivity function, and the same function says how much of each disturbance the loop removes. Reshaping the controller with a lead and a lag compensator moves the loop away from -1, removes more of a steady drag and stops the derivative from feeding sensor noise to the motor.

The robot's heading loop is a PD controller (G-2962), a torque from the heading error and its rate of change. RO-060 tuned it to Kp = 0.2 N · m/rad and Kd = 0.283 N · m · s/rad, plus an integral gain Ki = 0.05 against a dragging gearbox. Here we study RO-061's lightly damped first try of the turn, Kp = 0.4 N · m/rad and Kd = 0.08 N · m · s/rad with no integral action, because its sensitivity peak and its noise show what loop shaping fixes; Section 7.5 compares it with RO-060's tuned loop. The robot's yaw inertia is 0.1 kg · m^2. Three Notes have measured the first-try loop:

- its **Bode plot** (G-3023) shows a resonant peak of 8.1 dB at 1.92 rad/s;
- its **Nyquist plot** (G-3040), the frequency response of the loop cut open (controller times robot) drawn as a curve, stays clear of the critical point -1, so the loop is stable;
- its **phase margin** (G-3045) is 22.6°, its **gain margin** (G-3044) is infinite and its **delay margin** (G-3046) is 0.19 s.

An infinite gain margin suggests a safe loop. Figure 1 tests the loop in use. At t = 1 s the worn right gearbox of RO-060 starts to drag: a steady **torque** (G-2808), a turning effect, of 0.02 N · m. Watch the top panel: the loop as modelled rings and settles 0.05 rad (2.9°) off course. Then the bottom panel: the real robot carries a payload that raises its inertia to 0.15 kg · m^2, and its motor driver adds a delay of 0.25 s; the same drag starts a swing that keeps growing, 0.26 rad by 20 s.

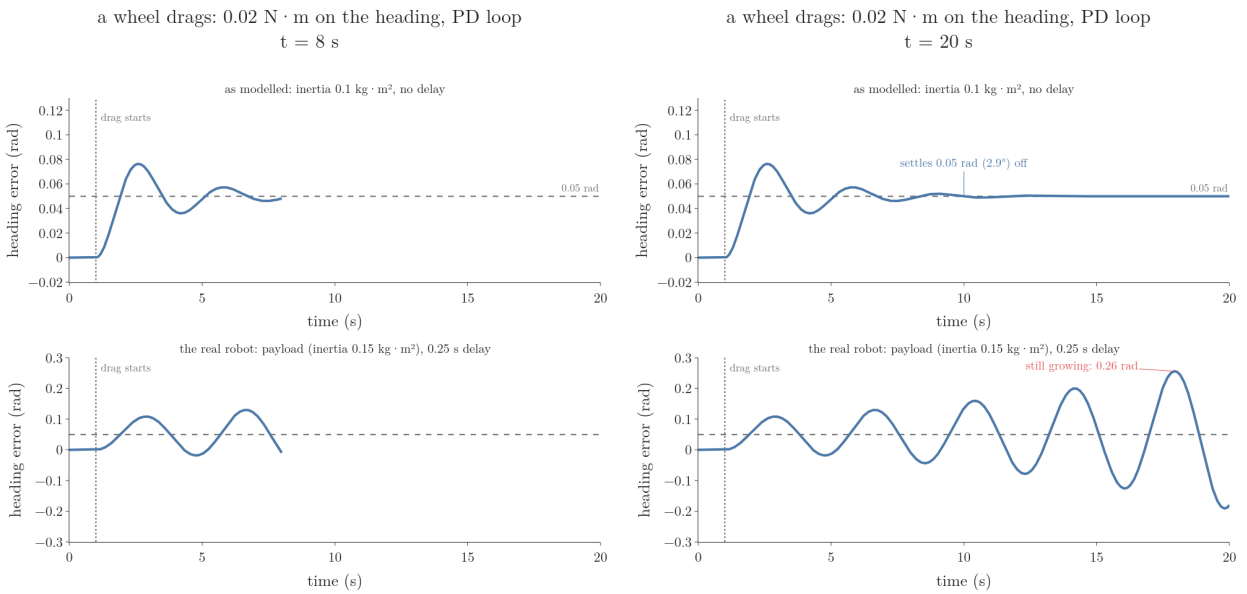


Figure 1: The heading loop under PD control (Kp = 0.4, Kd = 0.08) when a 0.02 N · m drag starts at t = 1 s, drawn one second per frame. Top: as modelled (inertia 0.1 kg · m^2, no delay), the heading error rings, peaks at 0.076 rad and settles at 0.05 rad. Bottom: the real robot (inertia 0.15 kg · m^2, 0.25 s delay); the swing grows to 0.26 rad by 20 s. The last frame holds both

The loop has four problems, each with a number:

1. **The drag:** a steady 0.02 N · m pull leaves the heading 0.05 rad off (RO-060's tuned PD, with its smaller Kp of 0.2, leaves 0.1 rad).
2. **How close to −1:** the Nyquist curve passes only 0.392 from −1, so a small change of the loop can reach it.
3. **The noise:** the derivative multiplies noise on the heading reading by Kd times its frequency, so noise at 100 rad/s reaches the motor 8.0 N · m per rad of noise, and faster noise more.
4. **The real robot:** the payload and the 0.25 s delay make the loop unstable, as the delay margin of 0.19 s warned.

The opening question of this Note: **how do we read off how close the loop is to instability, how it treats drags and sensor noise at each frequency, and how do we reshape the controller so that the drag leaves less than 0.01 rad, the noise reaching the motor stops growing, and the real robot settles?** The Note answers it in steps:

- the closest distance to −1 and the **sensitivity function** (G-3050) built from it (Section 2);
- the same function as the path of a disturbance through the loop (Section 3);
- the path of sensor noise, the **complementary sensitivity function** (G-3051) (Section 4);

- all four paths together, the **Gang of Four** (G-3054), with the drag and the noise at the motor (Section 5);
- whether the loop stays stable on the real robot, **robust stability** (G-3058) (Section 6);
- the fix, **loop shaping** (G-3060) with lead and lag compensators (Section 7);
- the limit that holds for every controller, the **waterbed effect** (G-3059) (Section 8).

2. How near the loop passes to the critical point: the sensitivity function

> **Key point:** The distance from -1 to the Nyquist curve at each frequency is the size of $1 + L$. Its reciprocal is the gain of the sensitivity function S, and the peak of that gain, Ms, is one over the closest distance: 2.55 for the heading loop.

2.1 Margins measure two directions only

The second problem of the opening, how close the loop is to -1, needs a number that the margins do not give. The Nyquist curve is the frequency response of the loop cut open, the controller times the robot, drawn as points in the plane of complex numbers, and -1 is the **critical point** (G-3036), where a sine sent round the loop comes back the same size and inverted, so it can keep itself going. A gain margin measures how far the curve can be stretched before it reaches -1; a phase margin measures how far it can be turned (FBS §10.3). A real change, such as the payload of Figure 1, stretches and turns the curve at once, so the curve can reach -1 by a route that neither margin measures (FBS §10.3, Example 10.8).

The heading loop shows the gap. Its gain margin is infinite: no gain increase alone can make the loop unstable. Yet the curve passes 0.392 from -1 (Figure 2), and a model error that moves the curve by more than that, near the closest frequency, can reach -1. A single number that covers every direction is the shortest distance from -1 to the curve.

2.2 The distance from the critical point and its reciprocal

At each frequency the loop's response is a point on the Nyquist curve, and we measure how far that point is from -1. Figure 2 runs the frequency up and draws the red line from -1 to the point. Watch the line shorten until about 2 rad/s and lengthen again; its shortest length, 0.392 at 2.085 rad/s, is the closest approach.

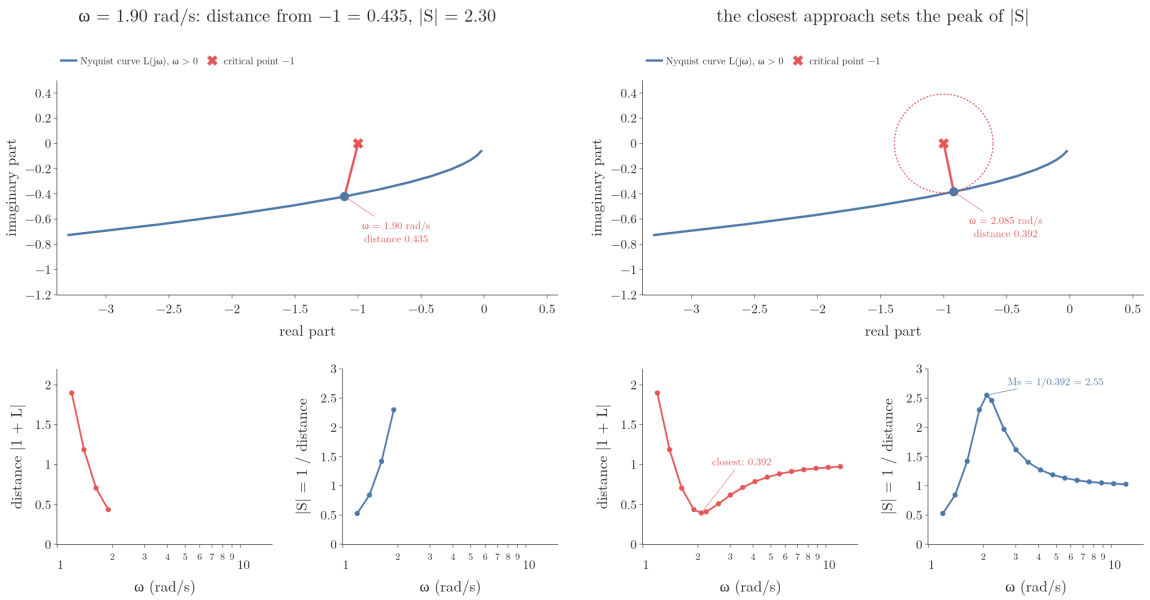


Figure 2: The sensitivity function read off the Nyquist plot of the PD heading loop, L(s) = (0.8 s + 4)/s^2, one frequency per frame from 1.2 to 12 rad/s. Top: the red line from −1 to the point L(jω) on the Nyquist curve ($\omega > 0$ only). Bottom left: the line's length |1 + L| against ω. Bottom right: its reciprocal, the gain |S| of the sensitivity function. The last frame marks the closest approach, 0.392 at 2.085 rad/s, and the peak Ms = 1/0.392 = 2.55, and draws the circle of radius 0.392 round −1

The symbols, at one frequency ω:

- L is the **loop transfer function** (G-3035), the controller times the plant (no unit); at the frequency ω it is a complex number, a point of the Nyquist curve;
- j is the imaginary unit, with $j^2 = -1$;
- $|z|$ is the size of a complex number z, its distance from 0;
- s is the Laplace variable of the transfer functions (G-2999), the ratios of output to input; putting $s = j\omega$ gives the response to a sine of frequency ω;
- S is the sensitivity function, defined below (no unit).

For the heading loop:

$$L(s) = \frac{0.8\,s + 4}{s^2}$$

The distance from −1 to the point L is the size of $L - (-1)$, which is $1 + L$. At the closest frequency, 2.085 rad/s, the loop's value is (notebook):

$$L = -0.920 - 0.384j$$

The distance:

$$1 + L = 0.080 - 0.384j$$

$$|1 + L| = \sqrt{0.080^2 + 0.384^2}$$

$$= \sqrt{0.0064 + 0.1475}$$

$$= \sqrt{0.1539} = 0.392$$

Its reciprocal:

$$|S| = 1/0.392 = 2.55$$

The same steps in symbols:

$$\text{distance} = |L - (-1)| = |1 + L|$$

$$S = \frac{1}{1 + L}$$

$$|S| = \frac{1}{|1 + L|}$$

The standard terms (FBS §12.1, §10.3):

- the transfer function S above, one over one plus the loop, is the **sensitivity function** (G-3050);
- the largest value of $|S|$ over all frequencies is the **maximum sensitivity** (G-3052), M_s, here 2.55;
- the shortest distance from -1 to the Nyquist curve is written s_m, here 0.392.

Because the gain of S is one over the distance, the peak of the gain sits where the distance is smallest:

$$M_s = \frac{1}{s_m} = \frac{1}{0.392} = 2.55$$

So Ms is the reciprocal of the closest distance, not the distance itself: a large Ms means a curve that passes close to -1. For the heading loop the phase margin of 22.6° gives nearly the same information, because the closest point lies almost at the **gain crossover frequency** (G-3042), the frequency where the loop's gain is 1, where the chord from -1 to the curve has the length (FBS eq. 10.7; notebook):

$$2\sin(22.6^\circ/2) = 0.392$$

Reasonable designs keep the shortest distance s_m between 0.5 and 0.8, that is Ms between about 1.25 and 2 (FBS §10.3). With 2.55 the heading loop is outside that range. Ms measures the closeness to instability, and the same S also says what the loop does to a push from outside, which the next section shows.

3. Disturbances: the sensitivity function from the block diagram

> **Key point:** A disturbance that moves the heading directly reaches the measured heading multiplied by S. Where the gain of S is below 1 the loop shrinks the disturbance; where it is above 1, near 2.1 rad/s for the heading loop, the loop makes it larger.

3.1 A disturbance at the output passes through S

Section 2 found S as a distance on the Nyquist plot; the first problem of the opening, the drag, asks what the loop does to a push from outside. Figure 3 draws the heading loop with the three outside signals a robot meets (FBS §12.1):

- a load **disturbance** (G-2949) v, an unwanted outside effect that changes what the robot does, here a torque that enters beside the motor's torque, such as the 0.02 N · m drag;
- an output disturbance d, anything that moves the heading itself, such as a bump that knocks the robot round;
- **sensor noise** (G-2312) n, an error in the heading reading that does not move the robot at all.

A disturbance acts on the robot; noise changes only the reading, which is why the two enter the block diagram at different places. Each outside signal takes its own path through the loop, and this section follows d.

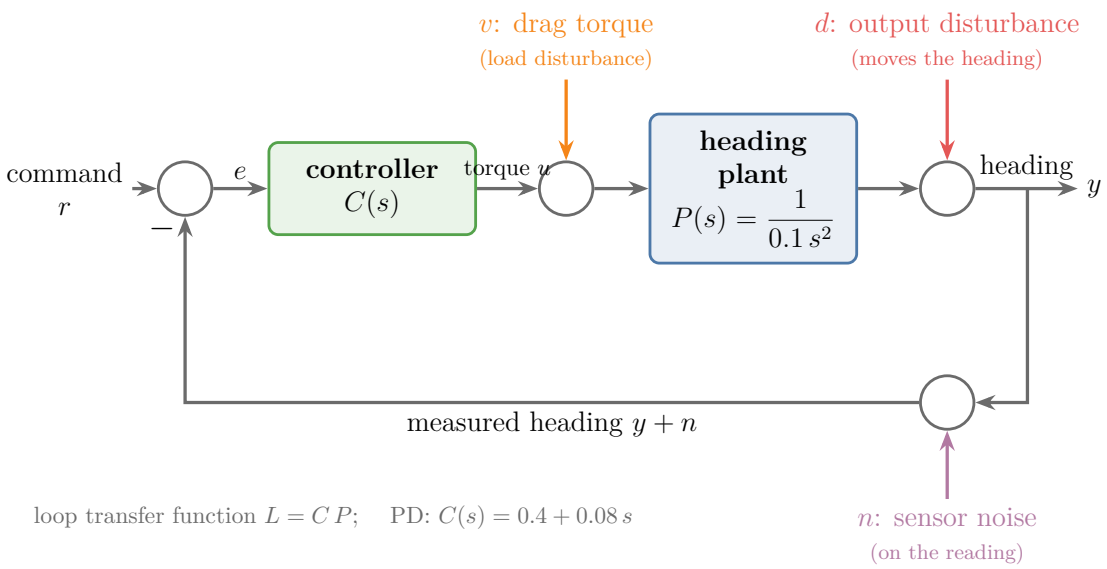


Figure 3: The heading loop with its outside inputs. The command r minus the measured heading gives the error e; the controller C(s) turns it into the motor torque u; the drag torque v adds to it; the heading plant P(s) = 1/(0.1 s^2) turns torque into heading; the output disturbance d adds to the heading y; the sensor noise n adds to the reading that goes back to the comparison. The loop transfer function is L = C P; for the PD controller C(s) = 0.4 + 0.08 s

To follow d round the loop at one frequency, we write each signal as the complex number of its sine, and every block multiplies it by its complex gain. With:

- y is the heading, as the complex number of its sine at the frequency ω (rad);
- d is the output disturbance, the same kind of number (rad);
- L is the loop transfer function at ω (no unit);
- the command r and the noise n are zero.

We take $\omega = 0.5$ rad/s, where the loop's value is (notebook):

$$L = -16 - 1.6j$$

The controller sees the error 0 − y, the loop turns it into L times that, and d adds on top. Our numbers first:

$$y = d + (-16 - 1.6j)(0 - y)$$

$$y = d + (16 + 1.6j)\, y$$

$$(1 - 16 - 1.6j)\, y = d$$

$$y = \frac{d}{-15 - 1.6j}$$

$$|y| = |d|/15.09 = 0.066\,|d|$$

The same steps in symbols:

$$y = d + L\,(0 - y)$$

$$y + L\,y = d$$

$$(1 + L)\, y = d$$

$$y = \frac{1}{1 + L}\, d = S\, d$$

The block diagram gives the same function as the Nyquist plot: the transfer function from an output disturbance to the heading is the sensitivity function (FBS §12.1). Lumping every push into one d at the output is a simplification; a push that enters elsewhere, such as the drag v, takes a path of its own, which Section 5 follows.

3.2 Disturbance attenuation: which frequencies shrink

Section 3.1 showed that a disturbance reaches the heading times S. The bottom right panel of Figure 2 therefore says, frequency by frequency, what the loop does to output disturbances:

- below 1.47 rad/s, $|S|$ is below 1: a slow disturbance is shrunk, to 0.066 of its size at 0.5 rad/s;
- above 1.47 rad/s, $|S|$ is above 1: a disturbance near 2.085 rad/s comes out 2.55 times larger than it went in;
- far above, $|S|$ falls back towards 1: the loop cannot react, and a fast disturbance passes almost unchanged.

Shrinking disturbances is **disturbance attenuation** (G-3057), and it happens where $|S| < 1$ (FBS §12.2). Feedback has a cost: the same loop that shrinks slow pushes amplifies those near its crossover, and the higher Ms, the larger the amplification. Disturbances are half of what reaches the loop; the other half is the noise in the reading, which takes a different path.

4. Sensor noise: the complementary sensitivity function

> **Key point:** Noise in the heading reading reaches the heading through T, the loop L divided by one plus L. Because S + T = 1 at every frequency, the loop cannot make both small at the same frequency: it rejects disturbances where L is large and noise where L is small.

4.1 Noise passes through T

The third problem of the opening is the noise. Noise n does not move the robot, but the controller cannot tell it from a real heading error, so it turns the robot to "correct" it (Figure 3). We follow n the way Section 3.1 followed d, with the disturbance and the command at zero, at the same 0.5 rad/s, where L is $-16 - 1.6$j:

$$y = (-16 - 1.6j)(0 - y - n)$$

$$(1 - 16 - 1.6j)\, y = (16 + 1.6j)\, n$$

$$y = -\frac{-16 - 1.6j}{-15 - 1.6j}\, n$$

$$|y| = (16.08/15.09)\, |n| = 1.066\, |n|$$

The same steps in symbols:

$$y = L\,(0 - y - n)$$

$$(1 + L)\, y = -L\, n$$

$$y = -\frac{L}{1 + L}\, n = -T\, n$$

The transfer function T just found, the loop divided by one plus the loop, is the **complementary sensitivity function** (G-3051) (FBS §12.1). At 0.5 rad/s the loop follows slow noise almost fully, 1.066 times its size: the robot physically turns to correct an error that does not exist. At high frequency L is small, so T is small and fast noise barely moves the heading. For a controller that acts only on the error e, as the lead-lag of Section 7 does, the same algebra shows that T also carries the command r to the heading (FBS §12.2).

4.2 S plus T equals one

At every frequency the two functions add to one, because their fractions share the bottom line. At 0.5 rad/s:

$$S + T = \frac{1}{-15 - 1.6j} + \frac{-16 - 1.6j}{-15 - 1.6j}$$

$$= \frac{-15 - 1.6j}{-15 - 1.6j} = 1$$

The same steps in symbols:

$$S + T = \frac{1}{1 + L} + \frac{L}{1 + L}$$

$$= \frac{1 + L}{1 + L} = 1$$

The sum is of complex numbers, not of gains: near the crossover S and T point in nearly opposite directions, so both gains can exceed 1 there. Figure 4 draws both gains for the heading loop. Read it from left to right:

- at low frequencies $|S|$ is small and $|T|$ is 1 (0 dB, the decibel value of a gain of 1): disturbances shrink, noise and commands pass;
- near 2 rad/s both peak, $M_s = 2.55$ and the largest gain of T, $M_t = 2.73$;
- at high frequencies $|T|$ falls and $|S|$ is 1: noise is ignored, disturbances pass.

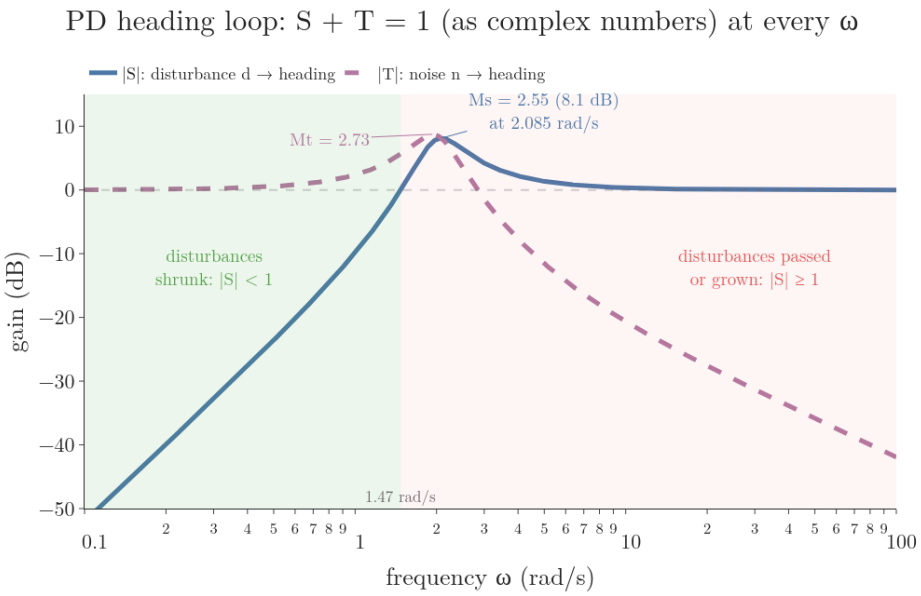


Figure 4: The gains of S (solid blue) and T (dashed purple) of the PD heading loop, in decibels against frequency. Green band, below 1.47 rad/s: $|S| < 1$, disturbances are shrunk. Red band, above: $|S| \geq 1$, disturbances pass or grow; the peak Ms = 2.55 (8.1 dB) is at 2.085 rad/s and Mt = 2.73 near 1.93 rad/s. T is about 0 dB at low frequency and falls at high frequency

Because S + T = 1, a loop cannot shrink disturbances and ignore noise at the same frequency; the designer chooses where each happens (FBS §12.1). Disturbances are usually slow and noise fast, so the usual choice is S small at low frequencies and T small at high ones. S and T cover the heading; the drag enters at the torque and the noise also reaches the motor, so two more paths are needed.

5. The Gang of Four: the drag and the noise at the motor

> **Key point:** Four transfer functions, S, T, PS and CS, describe how the outside signals reach the heading and the motor. PS sets the drag's offset, 1/Kp = 2.5 rad per N · m, and CS shows the derivative feeding fast noise to the motor, 8.0 N · m per rad at 100 rad/s and rising.

The first and third problems of the opening, the drag and the noise at the motor, are two paths that S and T do not cover. The drag v enters with the motor torque (Figure 3), and the noise n also moves the motor, which wears the actuator and can drive it into its limit (FBS §12.2); PX4's guide adds hot motors and shorter flights (PX4 filter tuning). Following each signal round the loop as in Section 3.1 gives four transfer functions (FBS §12.1), with P the plant, C the controller and L their product:

Name	Transfer function	Carries
S, sensitivity	$1/(1+L)$	output disturbance d to heading
T, complementary sensitivity	$L/(1+L)$	noise n (and command) to heading
PS	$P/(1+L)$	drag torque v to heading
CS	$C/(1+L)$	noise n to motor torque

Together they are the **Gang of Four** (G-3054) (FBS §12.1). Figure 5 draws them for the heading loop. Look at the two bottom panels.

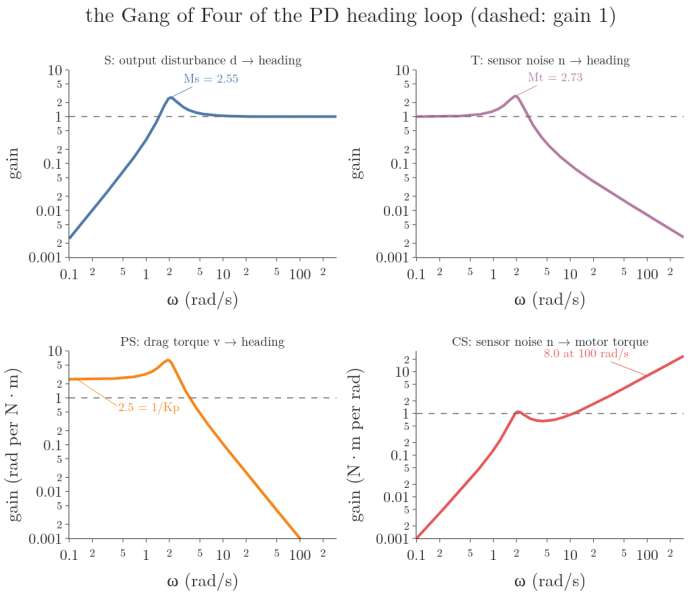


Figure 5: The Gang of Four of the PD heading loop (Kp = 0.4, Kd = 0.08, plant $1/(0.1\ s^2)$) against frequency, on log scales; dashed: gain 1. S: output disturbance to heading, peak Ms = 2.55. T: noise to heading, peak Mt = 2.73. PS: drag torque to heading, 2.5 rad per N · m at low frequency (= 1/Kp). CS: noise to motor torque, rising with frequency, 8.0 N · m per rad at 100 rad/s

The drag. A steady drag is a disturbance at zero frequency. The symbols:

- P is the heading plant (rad per N · m);
- C is the PD controller (N · m per rad);
- v is the drag torque, 0.02 N · m.

$$P(s) = \frac{1}{0.1\,s^2}$$

$$C(s) = 0.4 + 0.08\,s$$

Dividing the top and the bottom of PS by P:

$$PS = \frac{P}{1 + P\,C}$$

$$= \frac{1}{1/P + C}$$

At zero frequency the term 1/P, which is 0.1 times s squared, is 0, so only the controller's zero-frequency gain, Kp = 0.4, is left:

$$PS(0) = 1/0.4 = 2.5 \text{ rad per N} \cdot \text{m}$$

$$\text{offset} = 2.5 \times 0.02 = 0.05 \text{ rad}$$

The same in symbols:

$$PS(0) = \frac{1}{C(0)}$$

$$\text{offset} = \frac{v}{C(0)}$$

A check in plain mechanics: once the robot has stopped turning, the derivative term is zero, so the proportional torque alone balances the drag:

$$0.4 \times 0.05 = 0.02 \text{ N} \cdot \text{m}$$

The 0.05 rad of Figure 1 is this number. With RO-060's tuned gain Kp = 0.2 the same balance gives 0.1 rad, the 5.73° by which RO-060's robot stops short of its turn; RO-060 removes that offset with integral action, Ki = 0.05. To shrink the offset the controller needs more gain at low frequency; raising Kp would also move the crossover and the phase margin, so Section 7 adds gain at low frequencies only.

The noise at the motor. At high frequency L is small, so S is close to 1 and CS is close to the controller itself. For the PD controller the gain of C grows with frequency, because the derivative multiplies by Kd times the frequency:

$$|C(100j)| = |0.4 + 8j| = 8.0$$

Every 0.001 rad of ripple in the heading reading at 100 rad/s makes the motor torque swing by 0.008 N · m, 4 percent of the 0.2 N · m limit, and faster ripple more. The Gang of Four now shows all four problems of the opening on the PD loop; before changing the controller we need a test for the fourth one, the real robot.

6. Robustness to model error

> **Key point:** If the real plant differs from the model by a relative error δ, the loop stays stable as long as $|\delta|$ is smaller than $1/|\mathrm{T}|$ at every frequency. With the real robot's payload and delay, the PD loop fails that test between 1.6 and 2.4 rad/s, where its Nyquist curve passes close to -1.

6.1 The real robot as a disk round each point

The fourth problem of the opening is the real robot of Figure 1: the payload raises the inertia from 0.1 to 0.15 $\mathrm{kg \cdot m^2}$, and the motor driver adds a 0.25 s delay. A delay of T_d seconds multiplies a transfer function by e^{-sT_d}; at the frequency ω it keeps the gain and adds a lag of ω times T_d radians (a time delay (G-3012)). The symbols:

- P is the model, $1/(0.1\,s^2)$;
- P_{real} is the real robot, $e^{-0.25\,s}/(0.15\,s^2)$;
- δ is the relative model error, how far the real robot is from the model as a fraction of the model (no unit):

$$P_{\text{real}} = P\,(1+\delta)$$

Dividing the real robot by the model:

$$1+\delta = \frac{0.1}{0.15}\,e^{-0.25\,s}$$

$$\delta = 0.667\,e^{-0.25\,s} - 1$$

At 2.0 rad/s the delay lags by 0.5 rad, the frequency times the delay:

$$\delta = 0.667\,(\cos 0.5 - j \sin 0.5) - 1$$

$$= 0.585 - 0.320j - 1$$

$$= -0.415 - 0.320j$$

$$|\delta| = 0.524$$

The real loop is the model's loop times the same factor:

$$L_{\text{real}} = L\,(1+\delta) = L + L\,\delta$$

So each point L of the Nyquist curve, the frequency response of L drawn as points, moves by L times δ. Figure 6 draws, at five frequencies, a disk of radius $|L\,\delta|$ round the modelled point; the real point lies on its rim. Watch the disks at 1.7 and 2.0 rad/s: they cover -1, and the real curve (orange) has passed to the other side of -1.

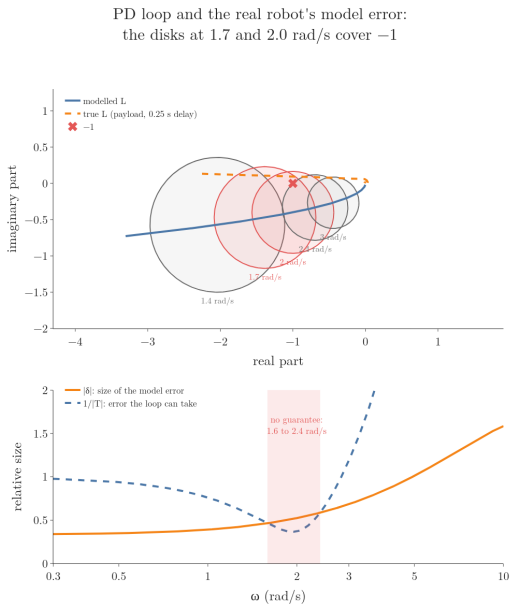


Figure 6: Top: the modelled Nyquist curve L of the PD heading loop (blue), the real robot's curve (orange, dashed: inertia 0.15 kg · m^2, 0.25 s delay) and, at 1.4, 1.7, 2.0, 2.4 and 3.0 rad/s, the disk of radius |L δ| round L(jω); the disks at 1.7 and 2.0 rad/s (red) cover -1. Bottom: the size of the model error $|\delta|$ (orange) against the bound 1/|T| (blue, dashed); between 1.6 and 2.4 rad/s (red band) $|\delta|$ is larger and the test fails

6.2 The test: the model error against one over T

A disk that stays clear of -1 at every frequency leaves the number of **encirclements** (G-3038) of -1, the net turns the curve makes round it, unchanged, so the real loop stays stable as the model's does (FBS §13.2). The disk stays clear when its radius is smaller than the distance from -1 to its centre. At 2.0 rad/s the loop's value is $-1 - 0.4$j (notebook):

$$|L| = \sqrt{1 + 0.16} = 1.077$$

$$\text{radius} = 1.077 \times 0.524 = 0.564$$

$$\text{distance} = |1 + L| = 0.400$$

The radius is larger than the distance, so the disk covers -1. The same test in symbols:

$$|L\,\delta| < |1 + L|$$

$$|\delta| < \frac{|1 + L|}{|L|} = \frac{1}{|T|}$$

A loop that passes this test at every frequency, for every model error of the expected size, has **robust stability** (G-3058) (FBS §13.2). The bottom panel of Figure 6 compares the two sides for the PD loop: at 2.0 rad/s, $|\delta| = 0.524$ against $1/|\mathrm{T}| = 0.371$, and the test fails between 1.6 and 2.4 rad/s, the band where the Nyquist curve passes close to -1. The test is cautious: it assumes the error points straight at -1, so passing it guarantees stability, while failing it only warns (FBS §13.2). Here the warning is right, since the real PD loop has a phase margin, the extra lag it could take at its crossover, of $-5.5°$, and its swing grows (Figure 1, notebook). RO-060's tuned PD passes the same test, its ratio of the two sides peaking at 0.544: on the real robot it keeps a phase margin of 41.9° at 2.00 rad/s, a gain margin of 3.05 and Ms = 1.73 (notebook), because its large derivative gain gave it a phase margin of 76.4° for the model error to use up.

The test says where robustness is needed: where T is large, near the crossover. All four problems now depend on the same thing, the shape of the loop, and the next section reshapes it.

7. Loop shaping: lead, lag and lead-lag

> **Key point:** Loop shaping chooses the controller so that |L| is high at low frequencies, crosses 1 gently and falls at high frequencies. A lead compensator adds phase at the crossover without unlimited high-frequency gain; a lag compensator adds gain at low frequencies; together, as a lead-lag, they fix all four problems of the heading loop.

7.1 The shape we want

Sections 3 to 6 showed what each frequency band of the loop must do for the robot. Because S, T, PS and CS are all built from L (and C), the bands become a wanted shape for the gain of L (FBS §12.3):

- **low frequencies:** $|L|$ large, so S is small and disturbances shrink; for the drag the controller's own gain at low frequency must be large, since the offset is the drag divided by that gain (Section 5);
- **the crossover:** $|L|$ passes through 1 gently, near -20 dB per **decade** (G-3024), a factor of 10 in frequency, because a steeper fall comes with a smaller phase margin, less room for extra lag (FBS §12.3, eq. 12.6);
- **high frequencies:** $|L|$ small and falling, so T and the noise gain CS are small; a controller whose gain falls at high frequencies has **high-frequency roll-off** (G-3064) (FBS §12.3).

Choosing the controller to give the loop this shape is **loop shaping** (G-3060) (FBS §12.3). Figure 7 lays the PD loop over the three bands. Look at the crossover: the PD loop falls at 37 dB per decade there (phase margin 22.6°); and at high frequencies it falls only 20 dB per decade, because the derivative keeps raising the controller's gain.

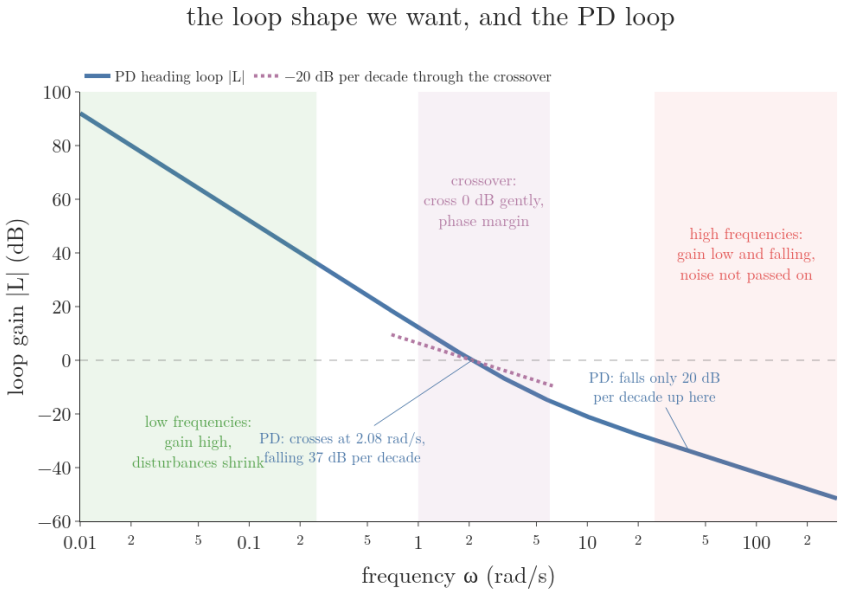


Figure 7: The gain of the PD heading loop |L| in dB (blue) against frequency, over the three bands of the wanted shape: low frequencies (green), gain high so disturbances shrink; the crossover (purple), cross 0 dB gently near −20 dB per decade (dotted guide) for phase margin; high frequencies (red), gain low and falling so noise is not passed on. The PD loop crosses at 2.08 rad/s falling 37 dB per decade and above 5 rad/s falls only 20 dB per decade

7.2 More derivative: Ms falls, noise rises

The first idea for the closeness to −1 is more derivative gain, the route RO-060's tuning took: Kd = 0.283 with Kp = 0.2. Figure 8 raises Kd step by step and draws the Nyquist curve, the frequency response of L as points, with the circle of radius s_m round −1, the gain of S, and the noise gain $|CS|$. Watch the curve swing away from −1 and the peak of S sink, while the noise gain at 100 rad/s climbs.

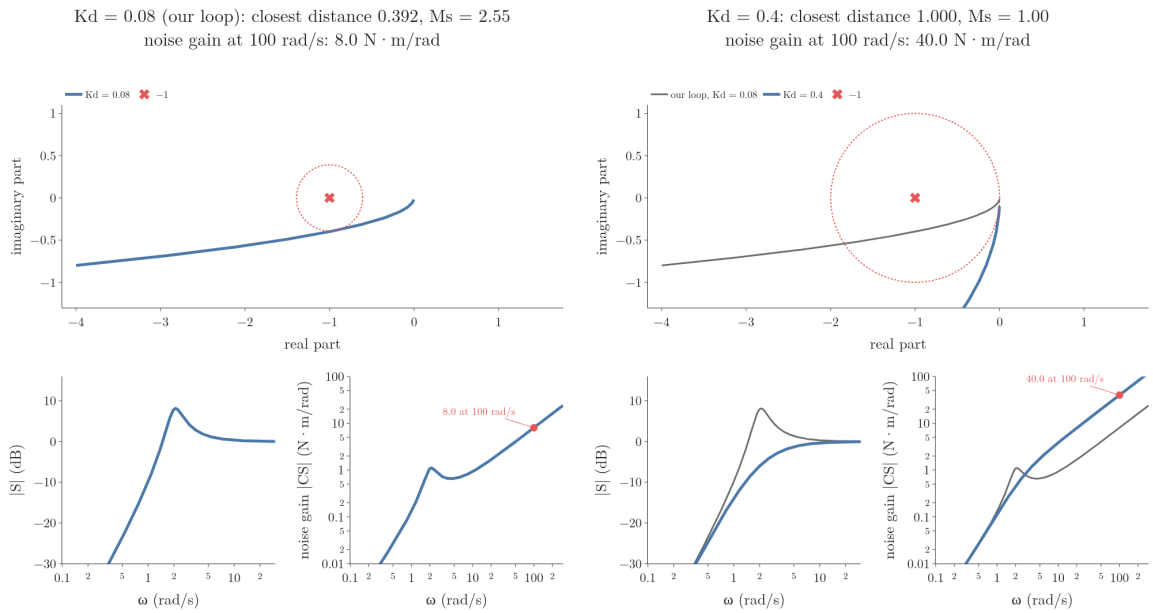


Figure 8: The PD heading loop with Kp = 0.4 and Kd = 0.04, 0.06, 0.08 (our loop), 0.12, 0.16, 0.2, 0.3 and 0.4, one value per frame; grey, our loop. Top: the Nyquist curve and the circle round −1 whose radius is the closest distance s_m. Bottom left: |S| in dB. Bottom right: the noise gain |CS| from heading noise to motor torque, in N · m per rad, with its value at 100 rad/s. Frame Kd = 0.08: s_m = 0.392, Ms = 2.55, noise 8.0. Last frame, Kd = 0.4: s_m = 1.0, Ms = 1.0, noise 40.0

The numbers from the notebook:

Kd (N · m · s/rad)	closest distance s_m	M_s	noise gain at 100 rad/s (N · m/rad)
0.04	0.199	5.03	4.0
0.08, our loop	0.392	2.55	8.0
0.16	0.733	1.36	16.0
0.4	1.000	1.00	40.0
RO-060's tuned loop: 0.283 (with Kp = 0.2)	1.000	1.00	28.3

The derivative trades one problem for another: Kd = 0.4 removes the closeness to −1 but sends five times more noise to the motor than Kd = 0.08, and RO-060's tuned loop, at the same end of the trade, sends 28.3 N · m per rad, 3.5 times ours. For this reason a practical derivative is filtered, so that the controller's gain stops rising at high frequencies (FBS §12.2). What is wanted is the derivative's phase near the crossover without its rising gain far above it.

7.3 The lead compensator: a derivative with a ceiling

A derivative adds phase because its gain rises with frequency: the rate of change of a sine is the same wave a quarter period earlier, +90°. A zero followed by a pole lets the gain rise only between the two, so the phase rises and falls back: a bump of phase, with the gain capped above the pole. The symbols:

- C is the compensator (N · m per rad);
- a is its zero, 0.8333 rad/s;
- b is its pole, 7.5 rad/s;
- k is its gain, 1.875 N · m/rad (found below).

$$C(s) = k\,\frac{s+a}{s+b}$$

With the pole above the zero, $a < b$, the compensator is a **lead compensator** (G-3061) (FBS §12.3, eq. 12.7). A lead is a PD controller with a filter on the derivative: below the pole it acts like PD, above it the gain stays at k (FBS §12.3).

How much phase, and where. The phase of C at the frequency ω is the angle of the zero's factor minus the angle of the pole's factor. The phase at 2.5 rad/s:

$$\arctan(2.5/0.8333) - \arctan(2.5/7.5)$$

$$= \arctan 3 - \arctan(1/3)$$

$$= 71.57^\circ - 18.43^\circ = 53.13^\circ$$

The bump is symmetric about the frequency whose square is the zero times the pole, 6.25, that is 2.5 rad/s. Two frequencies whose product is 6.25, such as 1 and 6.25 rad/s, give the same phase:

$$\arctan(1/0.8333) - \arctan(1/7.5)$$

$$= 50.19^\circ - 7.59^\circ = 42.6^\circ$$

$$\arctan(6.25/0.8333) - \arctan(6.25/7.5)$$

$$= 82.41^\circ - 39.81^\circ = 42.6^\circ$$

The same in symbols, for ω and the partner frequency ab/ω, using the rule for the angle of a reciprocal:

$$\arctan(1/x) = 90^\circ - \arctan x$$

The partner's phase:

$$\arctan \frac{ab/\omega}{a} - \arctan \frac{ab/\omega}{b}$$

$$= \arctan \frac{b}{\omega} - \arctan \frac{a}{\omega}$$

$$= \arctan \frac{\omega}{a} - \arctan \frac{\omega}{b}$$

So the single peak of the bump sits at $\sqrt{ab} = 2.5$ rad/s. We put it at the crossover we want, 2.5 rad/s, and choose k so that $|L| = 1$ there. The plant's gain at 2.5 rad/s:

$$|P(2.5j)| = 1/(0.1 \times 6.25) = 1.6$$

The lead's factors and the gain:

$$\frac{|2.5j + 0.8333|}{|2.5j + 7.5|} = \frac{2.635}{7.906} = 0.3333$$

$$k \times 0.3333 \times 1.6 = 1$$

$$k = 1/0.5333 = 1.875$$

Figure 9 grows the ratio of the pole to the zero from 1 to 9, keeping the bump centred on 2.5 rad/s and the crossover there. Watch the phase bump rise through the -180° line, the phase margin grow, and the Nyquist curve bend away from -1. At ratio 1 the controller is a plain gain and the curve runs straight through -1.

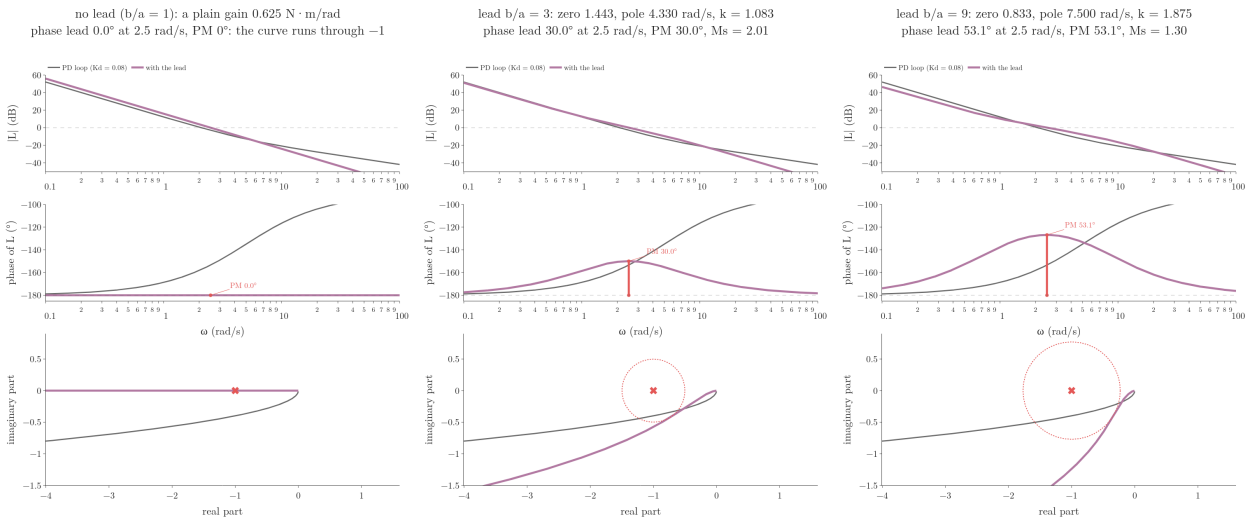


Figure 9: A lead compensator reshaping the heading loop. Its pole-to-zero ratio b/a grows 1, 1.5, 2, 3, 4.5, 6, 9, one per frame, centred on 2.5 rad/s with the gain set so the loop crosses 1 there; grey, the PD loop. Top: |L| in dB. Middle: the phase of L; the red bar from −180° at 2.5 rad/s is the phase margin. Bottom: the Nyquist curve and the circle round −1 of radius s_m. First frame, ratio 1: a plain gain 0.625, phase margin 0°, the curve runs through −1. Last frame, ratio 9 (zero 0.833, pole 7.5 rad/s, k = 1.875): phase lead 53.1°, phase margin 53.1°, Ms = 1.30

With the ratio 9, the lead loop has a phase margin of 53.1° and Ms = 1.30, against 22.6° and 2.55 for the PD loop, and its noise gain at 100 rad/s is 1.87 instead of 8.0 (notebook). One problem became worse: the lead's gain at zero frequency is

$$C(0) = 1.875 \times 0.8333/7.5 = 0.208$$

so the drag's offset grows from 0.05 rad:

$$\text{offset} = 0.02/0.208 = 0.096 \text{ rad}$$

The lead fixed the crossover and the noise; the low frequencies need their own fix.

7.4 The lag compensator: more gain at low frequencies

The drag's offset is the drag torque v (0.02 N · m) divided by $C(0)$, the controller's gain at zero frequency (Section 5), so it needs more controller gain at zero frequency without disturbing the crossover. A pole followed by a zero does that: the gain is high below the pole and back to 1 above the zero. The symbols:

- C_{lag} is the lag compensator (no unit);
- a_2 is its zero, 0.25 rad/s, a decade below the crossover;
- b_2 is its pole, 0.025 rad/s.

$$C_{\text{lag}}(s) = \frac{s + a_2}{s + b_2}$$

With the zero above the pole, $a_2 > b_2$, the compensator is a **lag compensator** (G-3062) (FBS §12.3). Its gain at zero frequency is the zero divided by the pole:

$$C_{\text{lag}}(0) = 0.25/0.025 = 10$$

At the crossover it adds a small phase lag:

$$\arctan(2.5/0.25) - \arctan(2.5/0.025)$$

$$= 84.29° - 89.43° = -5.14°$$

A PI controller is the special case $b_2 = 0$: its gain at zero frequency is infinite and the offset vanishes (FBS §12.3). RO-060 removed the gearbox's offset in exactly that way, with Ki = 0.05; the lag is the same idea seen in frequency terms, with a finite gain at zero frequency. Figure 10 adds the lag to the lead with the ratio a_2/b_2 growing from 1 to 10. Watch three things: $|L|$ lifting below 0.25 rad/s, $|S|$ sinking there, and the drag's offset falling in the bottom panel.

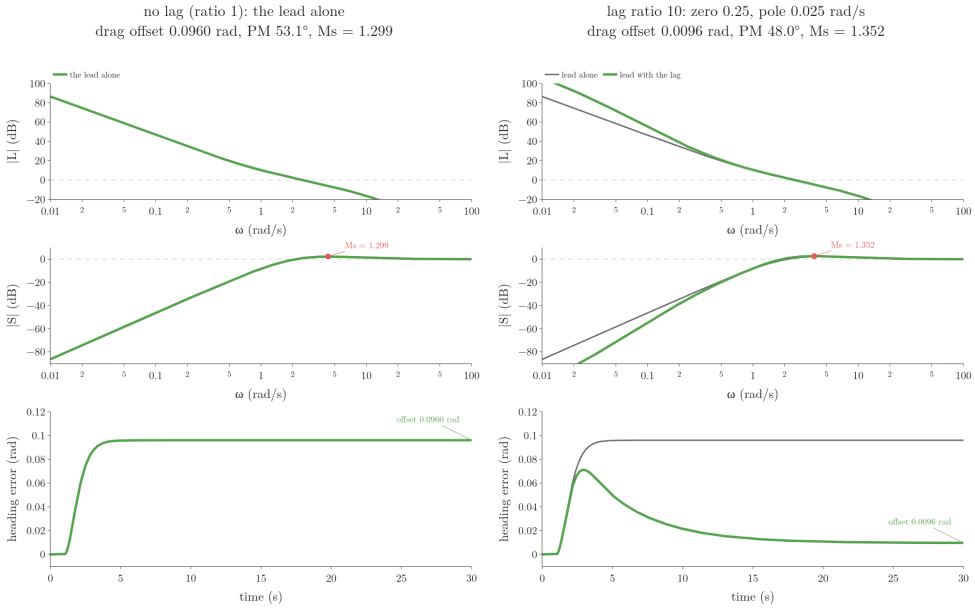


Figure 10: A lag compensator (s + 0.25)/(s + b_2) added to the lead, its ratio 0.25/b_2 growing 1, 2, 4, 6, 10, one per frame; grey, the lead alone (from the second frame on). Top: |L| in dB, lifted below 0.25 rad/s. Middle: |S| in dB, lower at low frequencies, its peak Ms rising from 1.299 to 1.352. Bottom: the heading error after the 0.02 N · m drag starts at t = 1 s; the offset it settles to falls from 0.096 rad (lead alone) to 0.0096 rad. Last frame, ratio 10: phase margin 48.0°, Ms = 1.352

The offset falls tenfold, from 0.096 to 0.0096 rad, and the phase margin drops by the lag's 5.1°, from 53.1° to 48.0°. The error takes about 20 s to decrease: the lag adds a slow closed-loop pole at -0.24 (a time constant of about 4 s, notebook), the price of acting only at low frequencies. The peak of S rose slightly, from 1.299 to 1.352, which Section 8 explains.

7.5 Lead-lag: the result on the robot

The lead and the lag in series form a **lead-lag compensator** (G-3063) (FBS §12.3):

$$C(s) = 1.875\,\frac{s + 0.8333}{s + 7.5}\,\frac{s + 0.25}{s + 0.025}$$

Its zero-frequency gain is the product of the two:

$$C(0) = 0.208 \times 10 = 2.08$$

$$\text{offset} = 0.02/2.08 = 0.0096 \text{ rad}$$

Figure 11 replays the opening problem with both controllers. Watch the bottom panel: on the real robot, with the payload and the 0.25 s delay, the PD swing grows while the lead-lag settles.

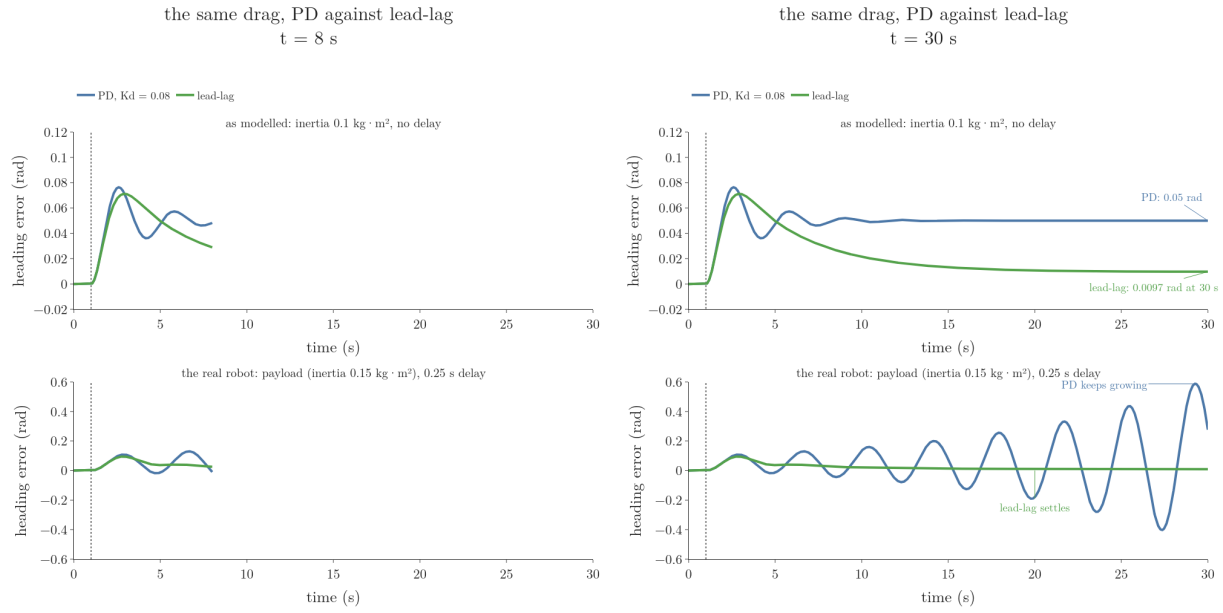


Figure 11: The opening problem replayed, PD (blue) against lead-lag (green), two seconds per frame; a 0.02 N · m drag starts at t = 1 s. Top, as modelled: the PD settles 0.05 rad off; the lead-lag peaks at 0.071 rad and creeps down, 0.0097 rad at 30 s on its way to 0.0096 rad. Bottom, the real robot (inertia 0.15 kg · m^2, 0.25 s delay): the PD swing keeps growing, to 0.59 rad by 30 s; the lead-lag peaks at 0.095 rad and settles. The last frame holds both

Figure 12 compares the frequency responses of our first-try PD, RO-060's tuned PD (Kp = 0.2, Kd = 0.283) and the lead-lag: the sensitivity, the noise gain at the motor, and the robustness test of Section 6.

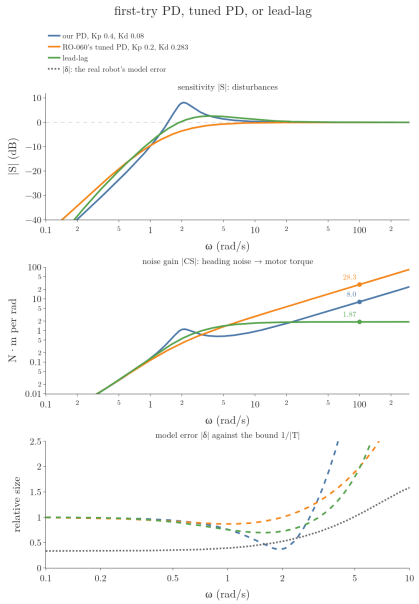


Figure 12: Top: |S| in dB of our first-try PD (blue, peak 2.55), RO-060's tuned PD with Kp = 0.2 and Kd = 0.283 (orange, never above 1) and the lead-lag (green, peak 1.35). Middle: the noise gain |CS| from heading noise to motor torque; at 100 rad/s 8.0, 28.3 and 1.87 N · m per rad; the lead-lag's stays flat at high frequency. Bottom: the real robot's model error $|\delta|$ (grey, dotted) against the bound 1/|T| of each loop (dashed, same colours); the bounds of the tuned PD and of the lead-lag stay above $|\delta|$ at every frequency, the first-try PD's dips below it

The four problems of the opening, before and after (notebook):

Problem	first-try PD	lead-lag	RO-060's tuned PD
drag offset	0.05 rad	0.0096 rad	0.1 rad (0 with Ki = 0.05)
closest distance to −1 (M_s)	0.392 (2.55)	0.740 (1.35)	1.000 (1.00)
noise gain at 100 rad/s	8.0 N · m/rad, rising	1.87 N · m/rad, flat	28.3 N · m/rad, rising
real robot (payload, 0.25 s delay)	grows, phase margin −5.5°	settles, phase margin 18.8°	settles, phase margin 41.9°

The lead-lag keeps the crossover near the first-try loop's (2.51 against 2.08 rad/s), so it does not trade speed for safety; its delay margin is 0.334 s against 0.19 s, and its robustness test passes at every frequency, the largest $|\delta|\,|T|$ being 0.743, below 1 (bottom panel of Figure

12). RO-060's tuned PD reaches its safety by a large derivative gain: crossover 2.91 rad/s, phase margin 76.4°, delay margin 0.458 s and |S| never above 1, and it keeps more margin on the real robot than the lead-lag: 41.9°, inside the usual 30° to 60° (FBS §10.3), against 18.8°, below it. The tuned PD's main cost is at the motor, 28.3 N · m per rad of noise at 100 rad/s, 15 times the lead-lag's 1.87, and without its integral term the drag leaves 0.1 rad. One cost remains, and it is not a matter of design skill.

8. The limit: the waterbed effect

> **Key point:** For a loop with no unstable poles whose gain falls at least as fast as $1/\omega^2$, the area under ln|S| plotted on a linear frequency axis is zero. Pushing |S| down at some frequencies pushes it up at others; no controller removes every disturbance at every frequency.

Section 7.4 pushed $|S|$ down at low frequencies with the lag and saw the peak rise from 1.299 to 1.352. The rise is not an accident of our design. A theorem of Bode states, for a loop with no poles in the right half-plane whose gain falls at least as fast as $1/\omega^2$ (FBS §14.2, Theorem 14.3):

- ln is the natural logarithm, so $\ln|S|$ is negative where disturbances shrink and positive where they grow;
- ω runs over all frequencies from 0 to infinity, on a linear scale.

$$\int_0^\infty \ln|S(j\omega)|\,d\omega = 0$$

The area where the loop shrinks disturbances equals the area where it amplifies them. Pushing $|S|$ down in one band lifts it in another, like pressing a waterbed: the **waterbed effect** (G-3059) (FBS §14.2). Figure 13 draws $\ln|S|$ for the lead and the lead-lag on a linear axis.

the waterbed: the area below zero equals the area above

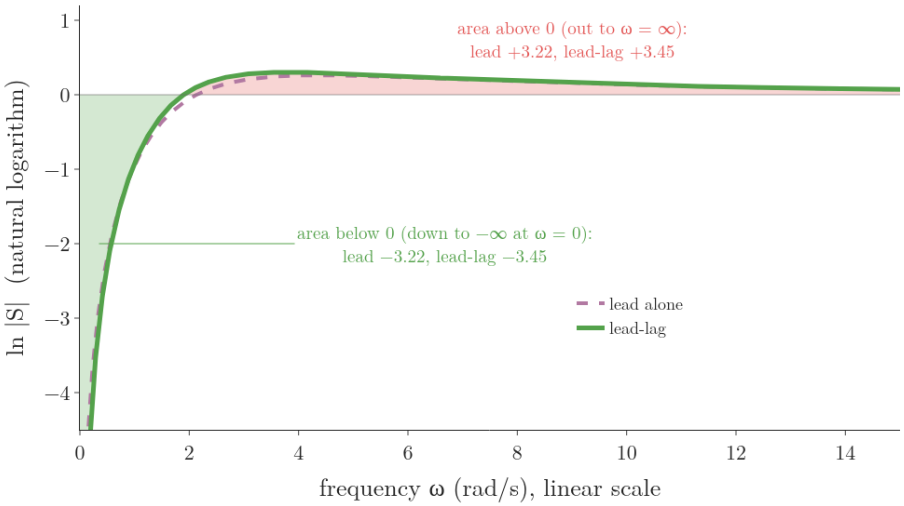


Figure 13: ln|S| of the heading loop against frequency on a linear axis, 0 to 15 rad/s: the lead alone (purple, dashed) and the lead-lag (green). Green shading: the area below zero, where disturbances shrink, down to minus infinity at $\omega = 0$; red shading: the area above zero, where they grow, which continues past 15 rad/s. Lead: areas −3.22 and +3.22; lead-lag: −3.45 and +3.45, of which 1.18 lies beyond 15 rad/s

The numbers (notebook):

Loop	area below zero	area above zero
lead alone	−3.22	+3.22
lead-lag	−3.45	+3.45

The lag deepened the green area by 0.23 and the red area grew by the same 0.23, as the peak of Figure 10 showed. On a linear axis the low-frequency band is narrow, so the lag's tenfold drop in the drag's offset costs only a small rise in the peak. For a loop with unstable poles the cost is larger: the total area is then π times the sum of those poles, a positive number (FBS §14.2). The modelled PD loop falls outside the theorem only because its controller's gain rises without bound, which is the noise problem of Section 5; a practical derivative is filtered (FBS §12.2), as PX4's is (Section 9), and then the loop's gain falls fast enough for the theorem to hold. The waterbed bounds what loop shaping can do; the next section shows where these trade-offs appear in real robots.

9. Where loop shaping runs in real stacks

> **Key point:** Apollo's control module wraps its speed and position controllers in lead-lag compensators; PX4 caps its derivative with filters and cuts a resonance with a notch; Autoware turns the derivative off and filters the speed error, because of noise.

The trades of this Note appear in the code of open robot and vehicle stacks:

- **Apollo** (Baidu's self-driving stack) has a lead-lag block, `leadlag_controller.cc`, in its control module. The block builds a compensator from three parameters, `alpha`, `beta` and `tau`: a gain `beta`, a zero at one over `tau` and a pole at one over `alpha` times `tau`, and turns it into a difference equation with **Tustin's method** (G-3113) at the control period, 0.01 s in the longitudinal controller's `controller_conf.pb.txt`. Its `lon_controller.cc` passes the output of the station (position) PID and of the speed PID through a lead-lag when `enable_reverse_leadlag_compensation` is on. The test file `leadlag_speed_conf.pb.txt` sets `alpha: 0.35`, `beta: 1.0`, `tau: 0.1`: a lead with its zero at 10 rad/s and its pole at 28.6 rad/s, giving 28.8° of phase at 16.9 rad/s (notebook). The shipped configuration leaves the block switched off, with `alpha` and `beta` 1 and `tau` 0, which passes signals unchanged (Apollo control).
- **PX4** (drone autopilot) filters the rate controller's derivative separately (`imu_gyro_parameters.yaml`): `IMU_DGYRO_CUTOFF`, 20 Hz by default, is a low-pass filter on the derivative term, "the most susceptible to noise", which lets the main gyro filter `IMU_GYRO_CUTOFF` (40 Hz) be set higher for less delay. The capped derivative is the lead's idea of Section 7.3. Its notch filter `IMU_GYRO_NF0_FRQ` removes a narrow band "to avoid feedback amplification of structural resonances"; a **notch filter** (G-3102) lowers Ms near one frequency without cutting the low-frequency gain. PX4's filter-tuning guide states the trade of Section 4: a higher gyro cutoff lowers the delay and allows higher P gains, but sends more noise to the motors (PX4 filter tuning).
- **Autoware** (self-driving stack) sets the derivative gain of its longitudinal PID to zero, `kd: 0.0` with `max_d_effort: 0.0`, in `autoware_pid_longitudinal_controller.param.yaml`; its README warns that "unintended noise may cause the output of the derivative term to be very large". The controller low-pass filters the speed error instead, `lpf_vel_error_gain: 0.9`: each new error counts 10 percent and the last filtered value 90 percent (`lowpass_filter.hpp`) (Autoware PID controller).

10. Summary

The controllers compared, on the heading loop (inertia 0.1 $\mathrm{kg \cdot m^2}$), with C the controller, s the Laplace variable, L the loop transfer function, S and T the sensitivity and complementary sensitivity functions and δ the relative model error:

Controller	$C(s)$	Phase margin	M_s	Drag offset	Noise gain at 100 rad/s	Why it matters
PD, first try (studied here)	$0.4 + 0.08\,s$	22.6°	2.55	0.05 rad	8.0, rising	close to -1, fails on the real robot
PD, RO-060's tuning	$0.2 + 0.283\,s$	76.4°	1.00	0.1 rad	28.3, rising	robust, 3.5 times our noise; needs Ki for the drag
lead	$1.875\,(s + 0.8333)/(s+ 7.5)$	53.1°	1.30	0.096 rad	1.87, flat	phase at crossover, noise capped
lead-lag	lead $\times$ $(s + 0.25)/(s + 0.025)$	48.0°	1.35	0.0096 rad	1.87, flat	fixes all four problems

- The heading loop's four problems are fixed by a lead-lag compensator: the drag's offset falls from 0.05 to 0.0096 rad, the closest distance to -1 grows from 0.392 to 0.740 (Ms from 2.55 to 1.35), the noise gain at 100 rad/s falls from 8.0 to 1.87 $\mathrm{N \cdot m}$ per rad and stops rising, and the real robot settles with a phase margin of 18.8° instead of a growing swing, because the loop was reshaped band by band.
- The **sensitivity function** (G-3050) S, one over one plus the loop L, has gain one over the distance from -1 to the Nyquist curve, so its peak **Ms** (G-3052) is the reciprocal of the shortest distance s_m from -1 to the curve, and one number covers every direction a model error can push.
- An output disturbance reaches the heading times S, so **disturbance attenuation** (G-3057) happens where $|\mathrm{S}| < 1$, below 1.47 rad/s for the PD loop, and the loop amplifies disturbances near its crossover by up to Ms.
- Sensor noise reaches the heading through the **complementary sensitivity function** (G-3051) T, the loop over one plus the loop, and S and T add to 1, so a loop cannot reject disturbances and ignore noise at the same frequency.
- The **Gang of Four** (G-3054) adds PS, the path from the drag torque to the heading, whose value at zero frequency, one over the controller's gain there, sets the drag's offset, and CS, the path from sensor noise to the motor torque, which tends to C at high

frequency, so a pure derivative feeds ever faster noise to the motor.

- **Robust stability** (G-3058) holds when the size of the relative model error δ stays below one over the gain of T at every frequency, because then no disk of possible real loops covers -1; the PD loop fails between 1.6 and 2.4 rad/s for the payload and the 0.25 s delay.
- **Loop shaping** (G-3060) asks for high loop gain at low frequencies, a gentle crossing near -20 dB per decade and **high-frequency roll-off** (G-3064), so each band does its job.
- RO-060's tuned PD gets its safety from a large derivative gain, so it is robust (phase margin 76.4°, 41.9° on the real robot against the lead-lag's 18.8°) but sends 15 times the lead-lag's noise to the motor; a lead caps the derivative's gain instead.
- A **lead compensator** (G-3061) is a derivative with a ceiling: its phase bump, 53.1° at the square root of zero times pole, adds phase margin while its gain stops rising above the pole.
- A **lag compensator** (G-3062) multiplies the low-frequency gain by the zero-to-pole ratio, 10 here, so the drag's offset falls tenfold for 5.1° of phase margin; PI is the case with the pole at zero.
- A **lead-lag compensator** (G-3063) combines both, so the crossover, the low frequencies and the high frequencies are each shaped by their own part.
- The **waterbed effect** (G-3059), Bode's integral, keeps the area under $\ln|S|$ at zero for a stable loop, so pushing disturbances down in one band lifts them in another; control design moves sensitivity, it cannot remove it.

The robot's heading loop now rejects a steady drag, keeps sensor noise off the motor and stays stable with a payload and a 0.25 s delay. Still missing: the lag removes the drag's offset only slowly, over about 20 s, and a large command still drives the torque into its 0.2 N · m limit; the next tools are feedforward, which adds the torque a known command needs before any error appears, anti-windup, which keeps an integrating part bounded at the torque limit, and cascaded loops, a fast inner loop under a slower outer one.

11. Sources

Built from

- Control System Lectures, "Understanding the Sensitivity Function", YouTube, https://www.youtube.com/watch?v=BAWdZvF1O40
- MATLAB, "Using Bode Plots, Part 1: Frequency Response Design Using Sensitivity Functions", https://www.mathworks.com/videos/using-bode-plots-closed-loop-systems-1-of-5-77057.html
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Scientists and Engineers, 2nd ed., Princeton University Press. §10.3 "Stability Margins" (eq. 10.7, Example 10.8, typical margins); §12.1 "Sensitivity Functions" (the Gang of Four, S + T = 1); §12.2 "Performance Specifications" (disturbance attenuation, Ms = 1/s_m, noise to the control signal); §12.3 "Feedback Design via Loop Shaping" (the loop shape, eq. 12.6, lead and lag compensation, eq. 12.7); §13.2 "Stability in the Presence of Uncertainty" (eq. 13.10); §14.2 "Bode's Integral Formula" (Theorem 14.3). Free: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-loopsyn_24Jul2020.pdf, fbs-robperf_24Jul2020.pdf, fbs-limits_24Jul2020.pdf, fbs-loopanal_24Jul2020.pdf (FBS)

Other references

- Apollo (Baidu), https://github.com/ApolloAuto/apollo, control module; the transfer function, the Tustin coefficients and the configured values were read from these files (Apollo control):
  - `modules/control/control_component/controller_task_base/common/leadlag_controller.cc`
  - `modules/control/control_component/proto/leadlag_conf.proto`
  - `modules/control/controllers/lon_based_pid_controller/lon_controller.cc` and `conf/controller_conf.pb.txt`
  - `modules/control/control_component/testdata/conf/leadlag_speed_conf.pb.txt` and `leadlag_station_conf.pb.txt`
- PX4 Autopilot, https://github.com/PX4/PX4-Autopilot (PX4 filter tuning):
  - `src/modules/sensors/vehicle_angular_velocity/imu_gyro_parameters.yaml` (IMU_GYRO_CUTOFF, IMU_DGYRO_CUTOFF, IMU_GYRO_NF0_FRQ)
  - `docs/en/config_mc/filter_tuning.md`, "MC Filter Tuning & Control Latency"
- Autoware Universe, https://github.com/autowarefoundation/autoware_universe (Autoware PID controller):
  - `control/autoware_pid_longitudinal_controller/config/autoware_pid_longitudinal_controller.param.yaml` and `README.md`
  - `control/autoware_pid_longitudinal_controller/include/autoware/pid_longitudinal_controller/lowpass_filter.hpp`

12. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Sensitivity function S (G-3050)	The transfer function $S = 1/(1 + L)$ of a feedback loop with loop transfer function L: it carries a disturbance at the output to the output, so the loop shrinks disturbances where $\|S\| < 1$, and its gain is one over the distance from -1 to the Nyquist curve.
Complementary sensitivity function T (G-3051)	The transfer function $T = L/(1 + L)$ of a feedback loop: it carries sensor noise (and, with error feedback, the command) to the output; since $S + T = 1$, a loop cannot make both small at the same frequency.
Gang of Four (G-3054)	The four transfer functions S, T, PS and CS that carry a feedback loop's outside signals (output disturbance, sensor noise, load disturbance) to the output and to the control signal; together they describe how the loop treats everything that enters it.
Robust stability (G-3058)	The property of a feedback loop of staying stable for every model error of an expected size; it holds when the relative model error $\|\delta\|$ stays below $1/\|T\|$ at every frequency, so no possible real loop reaches -1.
Loop shaping (G-3060)	Designing a controller by choosing the shape of the loop's gain curve: high gain at low frequencies to reject disturbances, a gentle crossing of 0 dB for phase margin, and low falling gain at high frequencies to ignore noise.
Waterbed effect (Bode's integral formula) (G-3059)	The rule that, for a loop with no unstable poles whose gain falls at least as fast as $1/\omega^2$, the area under $\ln \|S\|$ on a linear frequency axis is zero; pushing disturbances down at some frequencies pushes them up at others.
Maximum sensitivity M_s (G-3052)	The largest gain of the sensitivity function over all frequencies, equal to one over the shortest distance from -1 to the Nyquist curve; it is the worst amplification of a disturbance and a robustness measure, with 1.25 to 2 a usual target.

Term	Meaning
Disturbance attenuation (G-3057)	The shrinking of disturbances by feedback, which happens at the frequencies where the gain of the sensitivity function is below 1; near the crossover feedback amplifies disturbances instead.
High-frequency roll-off (G-3064)	A controller's gain falling (or at least no longer rising) at high frequencies, so that sensor noise is not passed on to the actuator; a lead's pole or a filter on the derivative provides it.
Lead compensator (G-3061)	A controller $k(s+a)/(s+b)$ with its zero below its pole ($a < b$): it adds a bump of phase, largest at $\sqrt{ab}$, to raise the phase margin at the crossover, while its gain stops rising above the pole; a PD controller with a filtered derivative.
Lag compensator (G-3062)	A controller $(s+a)/(s+b)$ with its pole below its zero ($b < a$): it multiplies the low-frequency gain by a/b to shrink steady offsets, at a small cost in phase at the crossover; PI control is the case $b = 0$.
Lead-lag compensator (G-3063)	A lead and a lag compensator in series, so the lead shapes the crossover and the high frequencies while the lag raises the low-frequency gain.
PD control (G-2962)	Feedback with a proportional part and a derivative part, such as K_p e + K_d de/dt; it acts like a spring and a damper, so it can bring a frictionless joint or heading to its target without swinging.
Bode plot (G-3023)	A pair of plots of a frequency response against the rate on a log scale: the gain in decibels and the phase in degrees; on it the pieces of a system add and the response is sketched with straight lines.
Nyquist plot (G-3040)	The loop transfer function drawn as the Nyquist contour is walked: mostly the frequency response, one point per frequency with the gain as length and the phase as angle, plus its mirror image; its turns round -1 decide closed-loop stability.

Term	Meaning
Phase margin (G-3045)	How much extra phase lag the loop can take at its gain crossover frequency before the closed loop becomes unstable, $180ř + \angle L(j\omega_{gc})$; typical designs keep 30° to 60°.
Gain margin (G-3044)	How many times the loop gain can grow before the closed loop becomes unstable, $1/\lvert L(j\omega_{pc})\rvert$ (often in dB); it says how far the Nyquist plot's crossing of the negative real axis is from -1.
Delay margin (G-3046)	The smallest extra time delay that makes the closed loop unstable, the phase margin in radians divided by the gain crossover frequency (0.19 s for the robot's heading loop); it says how late sensor readings or commands may arrive.
Torque (G-2808)	The turning effect of a force about a point: the force times its distance from the point, measured square to the force's line; comparing torques decides whether a pushed object tips.
Critical point (G-3036)	The point -1 of the complex plane: wherever the loop transfer function equals -1 the closed loop has a pole, so the Nyquist criterion counts turns round it and the margins measure the distance to it.
Loop transfer function $L(s)$ (G-3035)	The product of every transfer function a signal meets once round a feedback loop, such as controller times plant times delay, $L = CGe^{-sT_d}$; the closed loop's poles are where $L(s) = -1$, so stability and the margins are read off L alone.
Transfer function (control) (G-2999)	The number a linear system multiplies an exponential input e^(st) by, written as a function of s, such as G(s) = 1/(8s + 2) for the robot's speed and $10/s^2$ for its heading; it turns a differential equation into a fraction, so systems can be combined and read by algebra.
Gain crossover frequency ω_{gc} (G-3042)	The frequency at which the loop transfer function's gain is 1 (0 dB), such as 2.08 rad/s for the robot's heading loop; the phase margin and the delay margin are read there.

Term	Meaning
Disturbance (G-2949)	An unwanted outside effect that changes what the plant does, such as a ramp pulling a robot back with 0.785 N; it changes the true output, unlike sensor noise, which changes only the reading.
Sensor noise (G-2312)	Small random errors in a sensor's readings, so the same quantity measured twice gives slightly different numbers (a wall 2 m away reads 1.92 to 2.07 m); it is one reason a robot cannot take a reading at face value.
Time delay (dead time) (G-3012)	A wait of T seconds between a signal and its effect, such as a camera reporting the heading late, with transfer function e^(−sT); it has no poles or zeros but limits the gains a loop can use, here making the turn swing for ever near 0.32 s.
Encirclement (winding number) (G-3038)	One full turn of a closed curve round a point; the winding number is the net count, clockwise minus counter-clockwise, which the Nyquist criterion reads off the Nyquist plot round -1.
Decade (G-3024)	A factor of 10 in frequency, such as 0.25 to 2.5 rad/s; on a log axis every decade has the same width, and Bode-plot slopes are given per decade (-20 dB per decade for one pole).
Tustin's method (bilinear transform) (G-3113)	A discretisation rule that adds up each strip of an integral as a trapezoid joining its two samples, which amounts to replacing s by $(2/T)(z-1)/(z+1)$; it is the most accurate of the three simple rules and sends every stable continuous pole inside the unit circle.
Notch filter (G-3102)	A filter that blocks a narrow band around one frequency and passes the rest, $(s^2+\omega_0^2)/(s^2+(\omega_0/Q)s+\omega_0^2)$; it removes a known vibration, such as a frame's 10 Hz bending seen by the gyro, with far less lag at the loop's own frequencies than a low-pass.